Integrating to Infinity

Up to now, we've only dealt with integrals over a *finite* domain $[a, b]$. In fact, if you look back to when we proved FTC I, you'll see that it only deals with finite domains...

Definition Integrating over infinite domains (Part 1):

If $f(x)$ is continuous on $(-\infty, \infty)$ then

$$\int_a^\infty f(x)\,dx = \lim_{b\to\infty}\int_a^b f(x)\,dx \qquad \text{and} \qquad \int_{-\infty}^b f(x)\,dx = \lim_{a\to-\infty}\int_a^b f(x)\,dx.$$

These integrals are said to *converge* if the limits exist and are finite.

Examples

Compute the following integrals:

1. $\displaystyle\int_0^\infty e^{-2x}\,dx$

2. $\displaystyle\int_{-\infty}^{-1} \frac{1}{x^2}\,dx$

3. $\displaystyle\int_1^\infty \frac{1}{x}\,dx$

Definition Integrating over infinite domains (Part 2):

If $f(x)$ is continuous on $(-\infty, \infty)$ then

$$\int_{-\infty}^{\infty} f(x)\, dx = \int_{-\infty}^{a} f(x)\, dx + \int_{a}^{\infty} f(x)\, dx,$$

where a is any number. This integral only converges (i.e. only exists) when both integrals in the sum exist and are finite.

Example

Compute the integral $\displaystyle\int_{-\infty}^{\infty} \frac{1}{1+x^2}\, dx$.

Exercises

1. Calculate the following integrals, if they converge.

   (a) $\displaystyle\int_0^{\infty} \frac{e^x}{(e^x+1)^2} \mathrm{d}x$ (This is a substitution. Don't forget to change bounds!)

   (b) $\displaystyle\int_0^{\infty} xe^{-x} \mathrm{d}x$ (You may need L'Hopital's rule somewhere along the way...)

(c) $\displaystyle\int_0^{\infty} \frac{1}{(x+4)^2}\mathrm{d}x$

(d) $\displaystyle\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \csc^2(x)\mathrm{d}x$ (Hint: $\csc^2(x) = \frac{\sec^2(x)}{\tan^2(x)}$)

2. The probability that an an electrical component will fail by t months after its manufacture is given by

$$F(t) = \int_0^t \frac{1}{40} e^{-\frac{1}{40}x}\, dx$$

where t is given in months. Find the probability that a typical component will last:

(a) at most 20 months

(b) at least 40 months

(c) 40 to 50 months

3. Compare the following integrals to other integrals to see if they converge.

   (a) $\displaystyle\int_2^\infty \frac{1}{\sqrt{x^2-1}}\mathrm{d}x$
   (Hint: $\sqrt{x^2-1}$ is a little smaller than $\sqrt{x^2}=x$. What does that tell you about $\frac{1}{\sqrt{x^2-1}}$ compared to $\frac{1}{x}$?)

   (b) $\displaystyle\int_0^\infty \frac{1}{e^x+2^x}\mathrm{d}x$

   (c) $\displaystyle\int_1^\infty \frac{1+\sin x}{x^2}\mathrm{d}x$ (Hint: _____ $\leq \sin(x) \leq$ _____)

4. Find c such that $\displaystyle\int_{-\infty}^\infty f(t)=1$

$$f(t)=\begin{cases} cte^{-\frac{t}{2}} & t>0 \\ 0 & \text{otherwise} \end{cases}$$